

Dial Tool

A watch dial is a few things repeated round a centre. Dial Tool draws it that way: a list of rings, three hands and a date window, in millimetres, as clean vectors for print, etching and laser.

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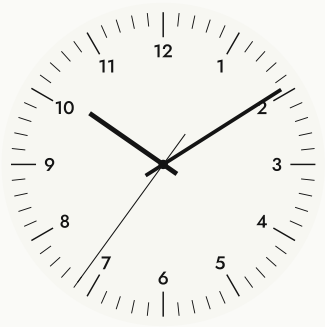
Pilot, type B · 16



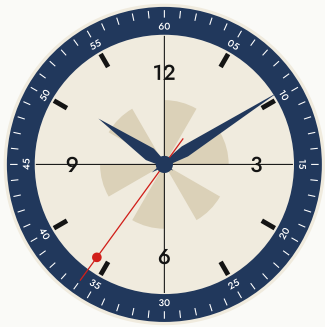
California · 17



Station clock · 18



Bauhaus · 19



Sector dial · 20



Gauge · 21

Start

Dial Tool is one Python file that runs inside DrawBot, a free drawing app for macOS.

INSTALL

Three steps, one file.

- 1 DrawBot 3.132 or newer, from [drawbot.com](#).
- 2 **dial.py** from [github.com/duztman/dial-tool](#). Branch **main** is the released version; **dev** is the newest work.
- 3 Open **dial.py** in DrawBot and press **⌘R**. The window opens.

UNIT

mm

Every length is in millimetres, drawn at true size. Radii are measured from the dial's centre.

ANGLE

0° = 12

Clock angles, growing clockwise. 90° is 3 o'clock, 180° is 6, -90° is 9.

SPEED

One optional package.

Knockouts and hollow shapes are cut by combining outlines. DrawBot's own routine is slow; **skia-pathops** is 2 to 14 times faster.

In DrawBot: Python → Install Python Packages → **skia-pathops** → Go. The Export log then starts with **shape combining: skia-pathops**.

MEMORY

Nothing to save.

Settings are stored as you work. **⌘R** again, or quitting DrawBot, keeps them.

UNIT

mm

Every length is in millimetres, drawn at true size. Radii are measured from the dial's centre.

ANGLE

0° = 12

Clock angles, growing clockwise. 90° is 3 o'clock, 180° is 6, -90° is 9.

SHAPE

Fills

Everything exported is a filled outline, never a stroke. A 0.12 mm tick is exactly 0.12 mm in Illustrator.

ORDER

Top first

The ring list is the stacking order. The top ring is drawn last, in front.

OUTPUT

Outlines

Numerals leave as outlines: no fonts needed to open or print. Live text is an option.

NUMBER

Slider, field and stepper show one value.

Length

1.00

Width · Ø

0.16

Drag for a rough value, type for an exact one, step for the last hundredth. Typing a value past the slider's end extends the slider.

TYPING

Fields do arithmetic.

1.2	sets 1.2
+0.5	adds 0.5
x0.5	halves
/3	a third

CHOICE

Popups, switches, checkboxes.

Kind

markers

Shape

bar

wedge

dot

file

Style

same

double

triangle

none

☐ Skip those instead

PREVIEW

Zoom and pan like a map.

Pinch	zoom about the cursor
⌘ + scroll	the same, with a mouse
Two fingers	pan
Fit	the whole page again

REFERENCE

Canvas or PDF.

The two previews draw the same scene and should look the same. If they don't, PDF is right – and worth a screenshot.

COLOUR

One flat colour per thing.

Colour

Outer radius

14.70

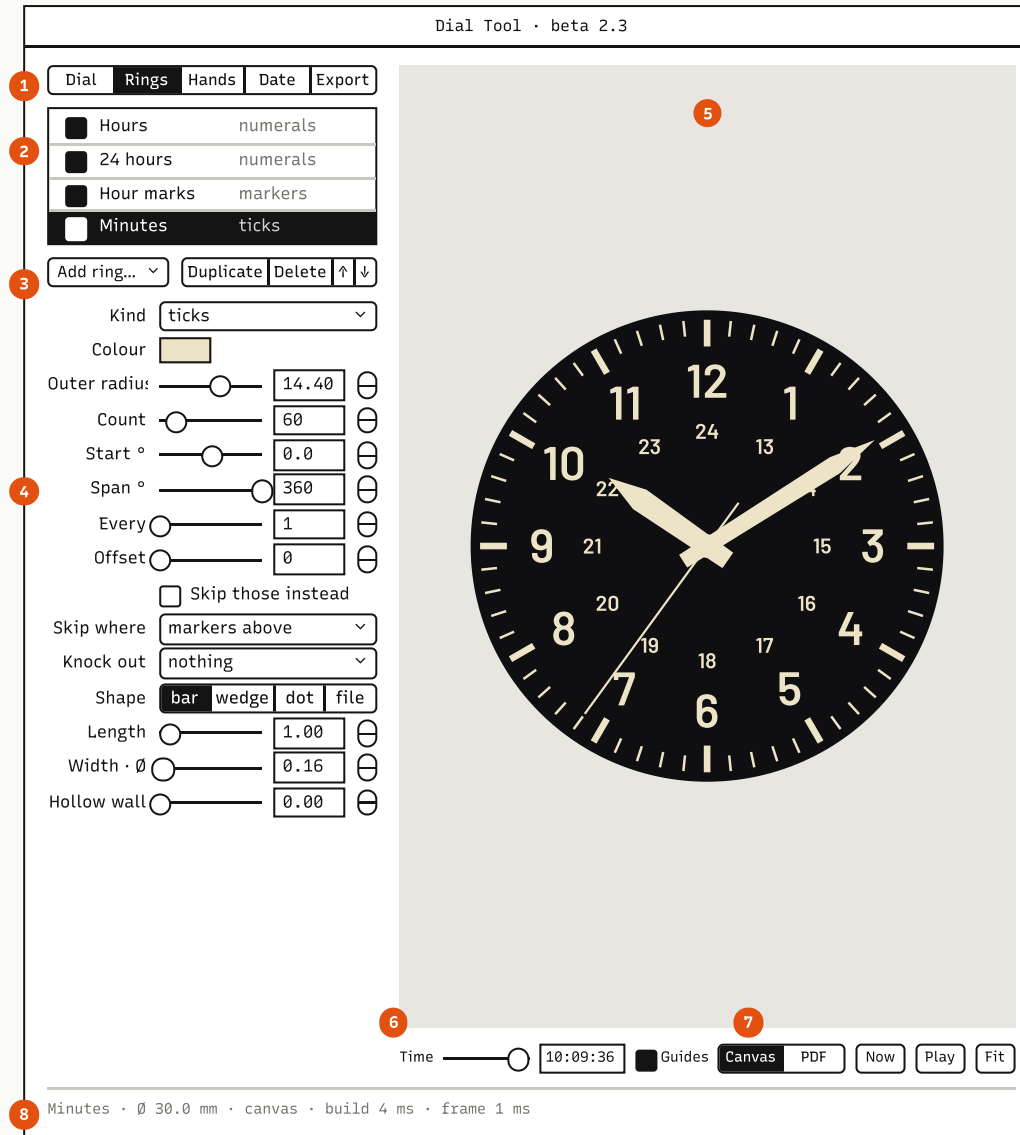
Band width

2.60

Rings, hands, cap, plate and date print each have a colour well.

The window

Settings on the left, the dial on the right, time underneath. Redrawn here; the real controls are standard macOS.



1 Sections

Dial, Rings, Hands, Date, Export. One is shown at a time.

2 Ring list

Top ring in front. The checkbox hides a ring; double-click a name to change it. The selected ring is the one the settings below edit.

3 Ring tools

Add ring... puts a new ring above the selected one. Duplicate, Delete, move up or down.

4 The selected ring's settings

One scrolling list. Rows that don't apply to the ring's kind are hidden, not greyed out.

5 Preview

Follows every change. With Guides on, the selected ring's edges are orange, the others blue.

6 Time

Drag, or type **10:09:36**. Now sets the clock's time; Play runs the hands.

7 Guides · Canvas · PDF · Fit

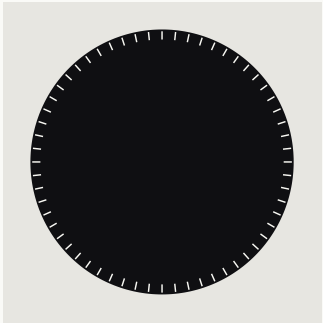
Guides are never exported. Canvas is the fast preview; PDF is DrawBot's own viewer, kept as the reference. Fit shows the whole page.

8 Status

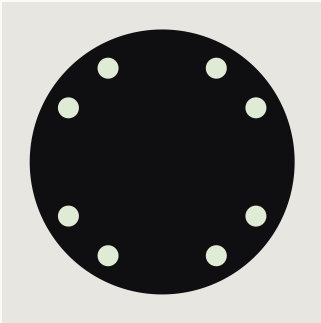
Ring, font, diameter; then how long the geometry (build) and the drawing (frame) took.

Rings

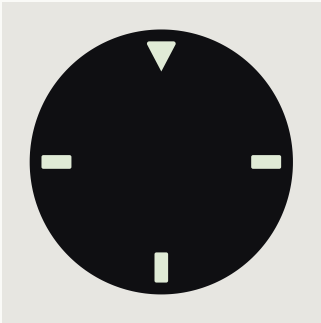
A ring repeats one thing round the centre, at its positions. A dial is a short list of them.



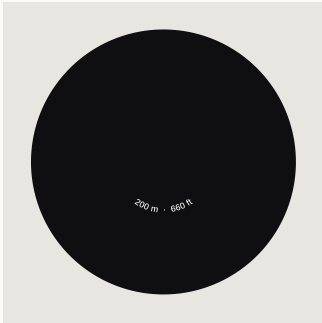
Minutes
ticks · Count 60



Dots
markers · Every 3, skip those instead



Bars · Triangle
markers · Count 4 · Count 1



Depth
numerals · Custom · on path



= the dial
plus hands and the date

THE IDEA

One dial, five rings.

The diver on page 14, taken apart. Each ring does one job; the list stacks them. Nothing here is special-cased: the triangle is a markers ring with a count of one.

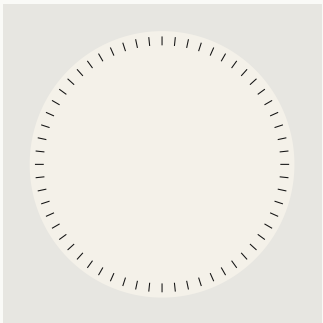
THE LIST

☐	Triangle	markers
☒	Bars 6 · 9	markers
☒	Dots	markers
☒	Depth	numerals
☒	Minutes	ticks

KIND

Four kinds, changeable at any time.

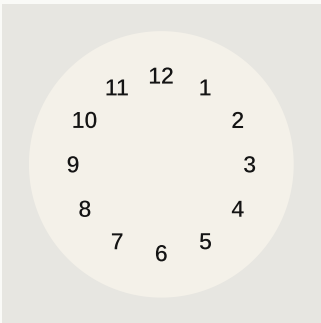
Ticks and **markers** repeat a shape. **Numerals** repeat a label. A **band** is colour between two radii. Every ring carries every setting, so changing a ring's kind loses nothing.



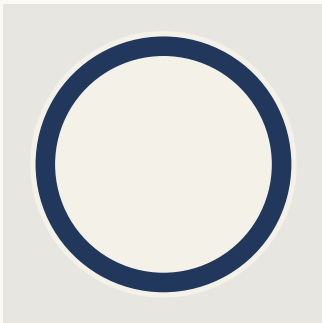
ticks
Count 60 · Length 1.0 · Width 0.12



markers
Count 12 · Length 2.8 · Width 0.7



numerals
Labels Arabic · Size 2.6



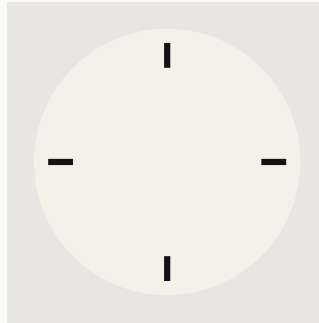
band
Radius 14.4 · Band width 2.2

Positions

Where a ring puts things. Six numbers describe almost any track, scale or set of hours.



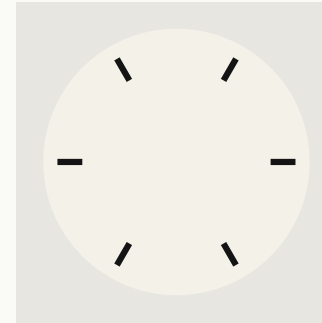
Count 12
Start 0° · Span 360°



Every 3
the cardinals



Every 3 · skip those instead
the other eight



Every 2 · Offset 1
the odd hours



Count 10 · Start -135° · Span 270°
an arc: both ends marked

COUNT · START · SPAN

A full turn spaces evenly. An arc marks both ends.

Count	10	⊖
Start °	-135.0	⊖
Span °	270	⊖

With Span 360° twelve positions sit 30° apart. With less, the first and the last position sit on the ends of the arc: Count 10 over 270° is a scale from 0 to 9.

EVERY · OFFSET · SKIP THOSE INSTEAD

Use some of the positions.

Every	3	⊖
Offset	0	⊖
☒ Skip those instead		

Every 3 keeps the cardinals. Tick **Skip those instead** and the same setting keeps the other eight. Offset moves the starting point.

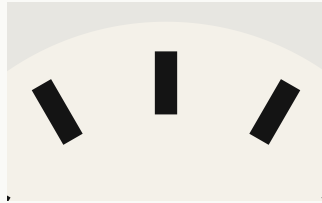
RADIUS

Outer edge.

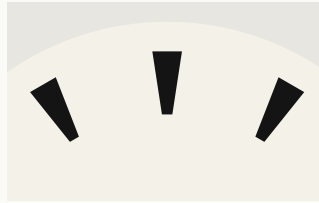
For ticks, markers and bands, Radius is the outer end. For numerals it is the centre of the text.

Marks

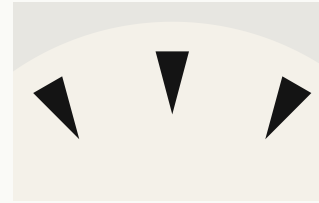
Ticks and markers share one set of shapes. Markers add a first position that can differ.



bar
Length 3.4 · Width 1.2



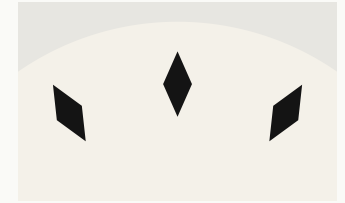
wedge
Inner width × 0.35



wedge
Inner width × 0



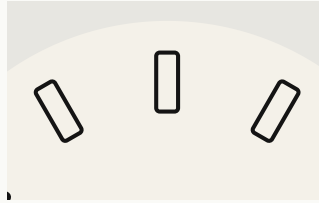
dot
Ø 2.0



file
your own SVG · PDF · AI



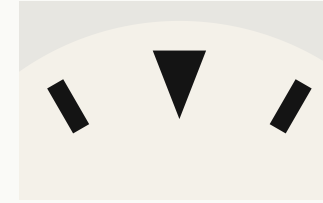
rounded
Round outer 0.6 · Round inner 0.2



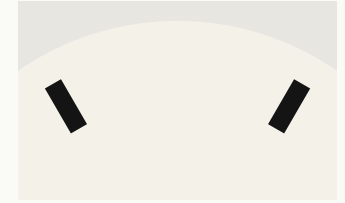
hollow
Hollow wall 0.22



12: double
Gap 0.4



12: triangle
Scale × 1.2



12: none

SHAPE

Bar, wedge, dot – or your own file.

Shape	bar	wedge	dot	file
Length				3.40
Width · Ø				1.20
Round outer				0.60
Round inner				0.20
Hollow wall				0.00

WEDGE

One number.

Inner width × sets the inner end as a share of the outer: 0 is a triangle, 1 a bar, more than 1 widens inward.

CORNERS

True fillets.

Round outer and Round inner are radii in mm, tangent on any angle. **Hollow wall** leaves an outline of that thickness.

12 0'CLOCK

The first position.

A markers ring can draw it the same, double (with a Gap), as a triangle, or not at all; Scale × sizes it.

Give way Bands

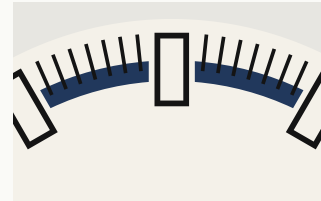
How a ring reacts to the rings above it, and how to lay colour.



stack
Skip where: nothing · Knock out: nothing



skip where
markers above



knock out
markers above · Clearance 0.25



knock out
markers & numerals above

GIVE WAY

Skip where:	nothing
Knock out:	markers above
Clearance:	0.25

SKIP WHERE

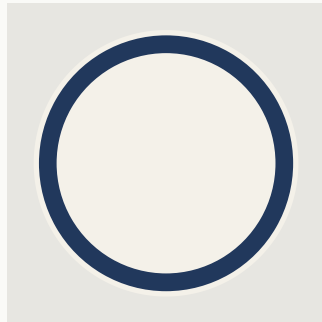
Leave out positions that are taken.

The ring drops its own positions wherever the chosen rings above already have a mark. Minute ticks that stop for the hour markers; fine ticks that stop for the minute ticks.

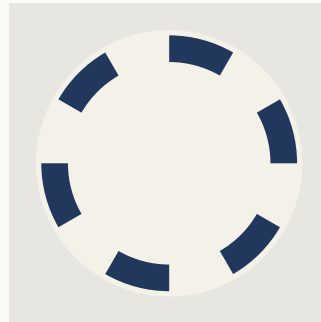
KNOCK OUT

Cut a gap round what is above.

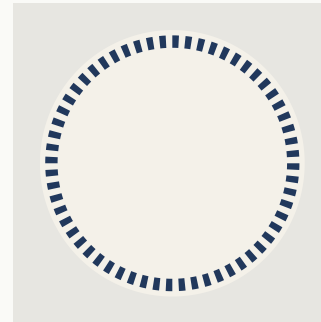
The ring is cut back by **Clearance** mm round the shapes above it: markers, markers and numerals, or everything. The date window always clears every ring, by its own Print clearance.



ring
Count 1 · Band width 2.0



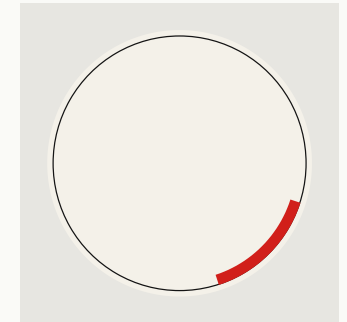
sectors
Count 12 · Every 2



fill
Count 60 · Fill 50 %



pies
Band width = radius



arc
Start 108° · Span 54°

BAND

Colour between Radius and Radius – Band width.

Count 1 is a full ring. More positions cut it into segments that tile the span; **Fill %** is how much of its step each segment covers.

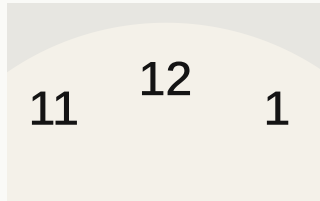
USES

Rails, zones, sectors.

Railroad rails are two bands 0.12 wide. A band as wide as its radius reaches the centre: with Every 2 that is a sector dial. A short span is a red zone.

Numerals

One label per position, from a built-in set, a count, or your own list.



Arabic



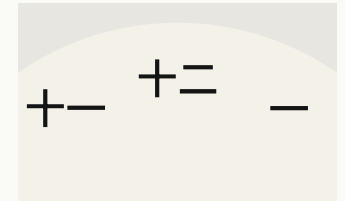
Roman • IIII



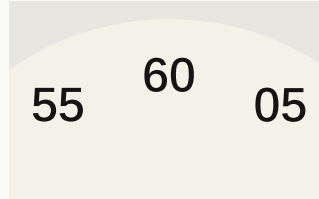
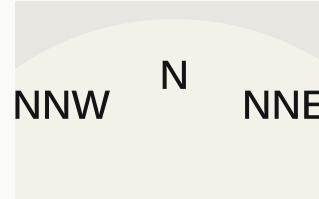
Roman • IV



Eastern Arabic



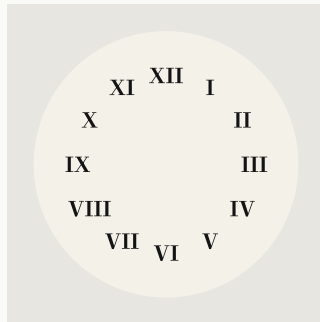
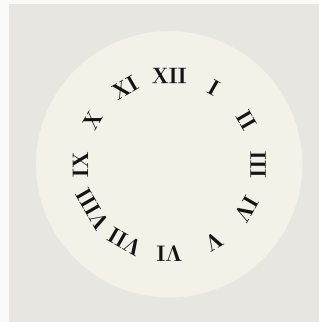
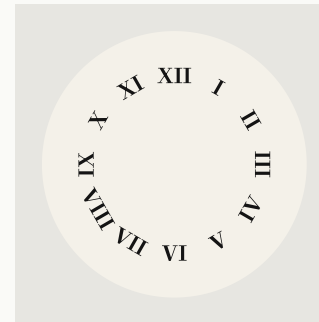
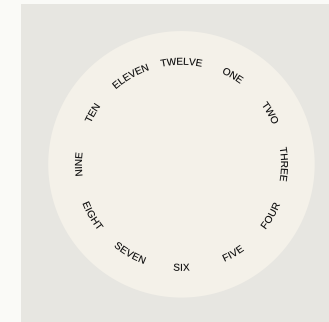
Chinese

Words
on pathNumbers
First 5 • Step 5 • Digits 2 • Start 30°Custom
N, NNE, , , E, ...

LABELS

Eight sets.

Labels	Numbers	v
First number	<input type="range"/>	5 0
Step	<input type="range"/>	5 0
Digits	<input type="range"/>	2 0
Start °	<input type="range"/>	30.0 0

upright
always levelradial
turns with the circleradial, auto-flip
never upside downon path
letters follow the circle

PLACEMENT

Four ways to sit.

On path sets each letter on the circle with the font's own spacing; below 3 and 9 it flips to read left to right.

NUMBERS

Labels that count.

From **First number** by **Step**, padded to **Digits**: 05, 10 ... 60 for a minute ring, 0, 10 ... 90 for a scale. Negative steps count down.

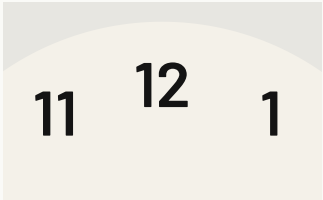
CUSTOM

Your own list, comma-separated.

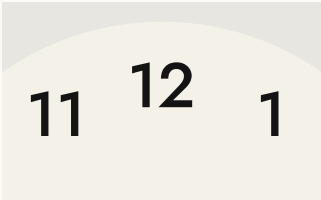
One entry per position. An empty entry leaves that position empty – which is how two rings can share one circle of hours.

Type Nudge

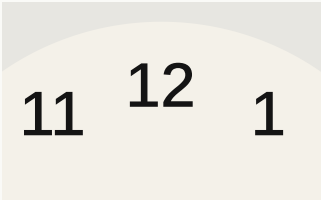
Each numerals ring has its own type. Each numeral can be corrected by eye.



DIN Alternate Bold



Futura Medium



Helvetica Neue Medium



Bodoni 72 Book

TYPE

What reaches the outlines.

Instance	a variable font's named instances
Variable axes	one slider per axis the font has
Features	OpenType features under the designer's own names: tabular or oldstyle figures, stylistic sets, slashed zero
Language	local forms, such as Turkish İ ı
Tracking	in 1/1000 em, as in Illustrator

NUDGE

Correct one numeral at a time.

Radius ±	in or out, mm
Angle ±°	round the dial
Rotate °	about its own centre
Size ×	larger or smaller

Stored per position. Reset this, or Reset all.

PER RING

Any installed font.

Family, then style. There is no fonts folder: the tool reads what macOS has.

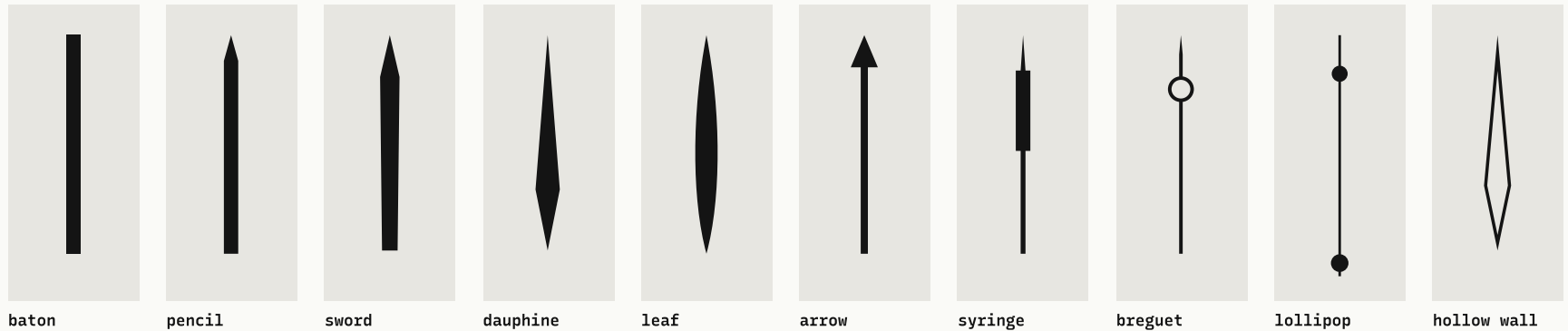
TYPICAL

Even the weight.

Two-digit numerals look heavier. Size × 0.92 on 10, 11 and 12 evens them; Radius -0.2 pulls a wide 12 in.

Hands

Hour, minute and seconds: each has its own shape, size and colour. Lengths are millimetres from the pivot, as movements specify them.



ONE HAND

The minute hand of the diver.

Shape	pencil	▼
Length	<input type="range"/>	12.40
Width	<input type="range"/>	1.50
Tail	<input type="range"/>	1.80
Tip length	<input type="range"/>	2.00
Counterweight Ø	<input type="range"/>	0.00
Hollow wall	<input type="range"/>	0.00

FEATURE

Head, ring, disc.

Arrow and syringe have a head or barrel width; Breguet and lollipop a ring or disc Ø and where it sits, in % of the length.

MOTION

Four beats.

sweep	smooth
quartz	1 per second
6 beats/s	21,600 vph
8 beats/s	28,800 vph

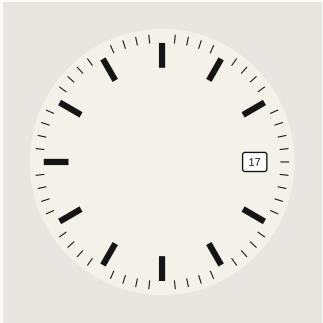
FILE

Your own hand.

Draw it pointing up with the pivot at the artboard's centre. **Fit file to length** scales it to the hand's Length; Width × stretches it.

Date window

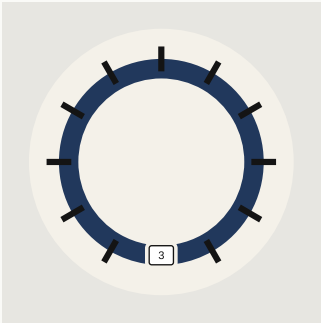
A hole in the plate that shows the date disc, with an optional printed frame.



at 3
Radius 10.5 · 2.6 × 2.0 · Frame 0.15



at 4:30
Corner 0.6 · Frame 0



at 6
Print clearance 0.4 cuts the band

THE WINDOW AT 3

☒ Date window

Position

Radius

Width

Height

Corner

Frame

Print clearance

Shows

Text % of height

CLEARANCE

Print keeps away from the hole.

Every ring is cut back by the frame plus **Print clearance**. Markers and numerals leave the date's position empty altogether, as on a real dial.

TYPE

The day uses the top numerals ring's font.

Shows is the day drawn on the disc; **Text % of height** sizes it. The frame and the day share the Print colour.

PRODUCTION

A cut line.

In a production mask the aperture is drawn as an outline to cut or drill to.

Classics

Nine dials drawn after well-known types, by eye. Approximations, not measured copies.

HOW TO READ THEM

Each page is a recipe.

Rings are listed top first. Only what differs from a newly added ring is printed; everything else is at its starting value. The values are the ones that drew the picture.

PRESETS

Load them.

Each dial is a settings file in [docs/guide/presets](#). Export → Settings → Load... opens it, ready to change.

FONTS

Named as on macOS.

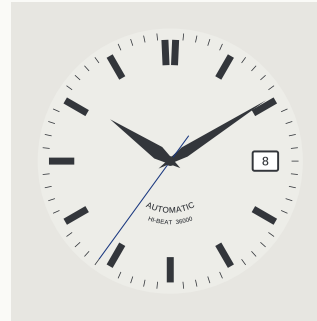
DIN Alternate, Futura, Helvetica Neue, Bodoni 72. These pages were rendered off-Mac with similar open fonts, so numerals differ slightly.



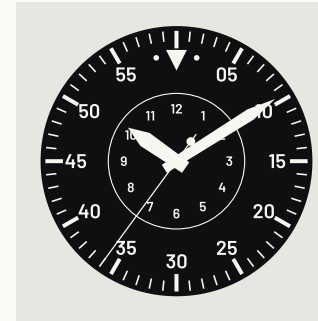
Field watch
page 13



Diver
page 14



Dress watch
page 15



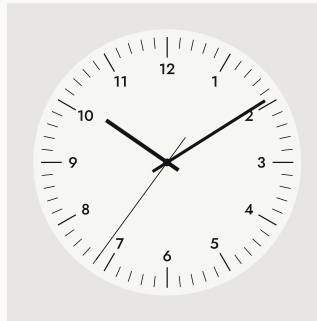
Pilot, type B
page 16



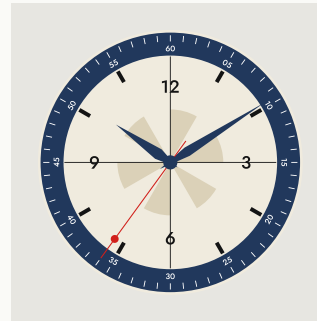
California
page 17



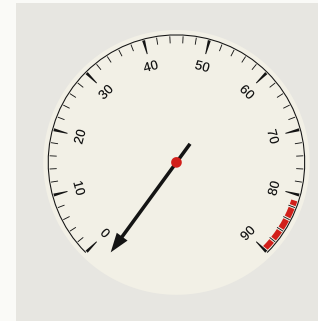
Station clock
page 18



Bauhaus
page 19



Sector dial
page 20



Gauge
page 21

Field watch

After the A-11 of the 1940s and its descendants: 12 and 24 hours, a plain minute track, nothing else.

LOOK FOR

The 24-hour ring is a second numerals ring with a Custom list. The minute ticks **skip** where the hour marks sit, so the two never overprint.

presets/field.json
Dial 30 mm ●



RINGS, TOP FIRST

☐ Hours	numerals
☒ 24 hours	numerals
☒ Hour marks	markers
☒ Minutes	ticks

SETTINGS THAT DIFFER FROM A NEW RING

Hours numerals	Radius 10.5 · Size 3 · Font DIN Alternate Bold ●
24 hours numerals	Radius 7.3 · Labels Custom · Custom 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 · Size 1.35 · Font DIN Alternate Bold ●
Hour marks markers	Radius 14.4 · Length 1.7 · Width 0.42 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 1 · Width 0.16 · Skip where markers above ●
Hour hand	Shape sword · Length 7.6 · Width 1.5 · Tail 1.6 · Tip length 2.4 ●
Minute hand	Shape sword · Length 12.6 · Width 1.2 · Tail 1.8 · Tip length 3 ●
Seconds hand	Shape baton · Length 13.2 · Width 0.14 · Tail 3.4 ●
Cap	Ø 1.5 ●

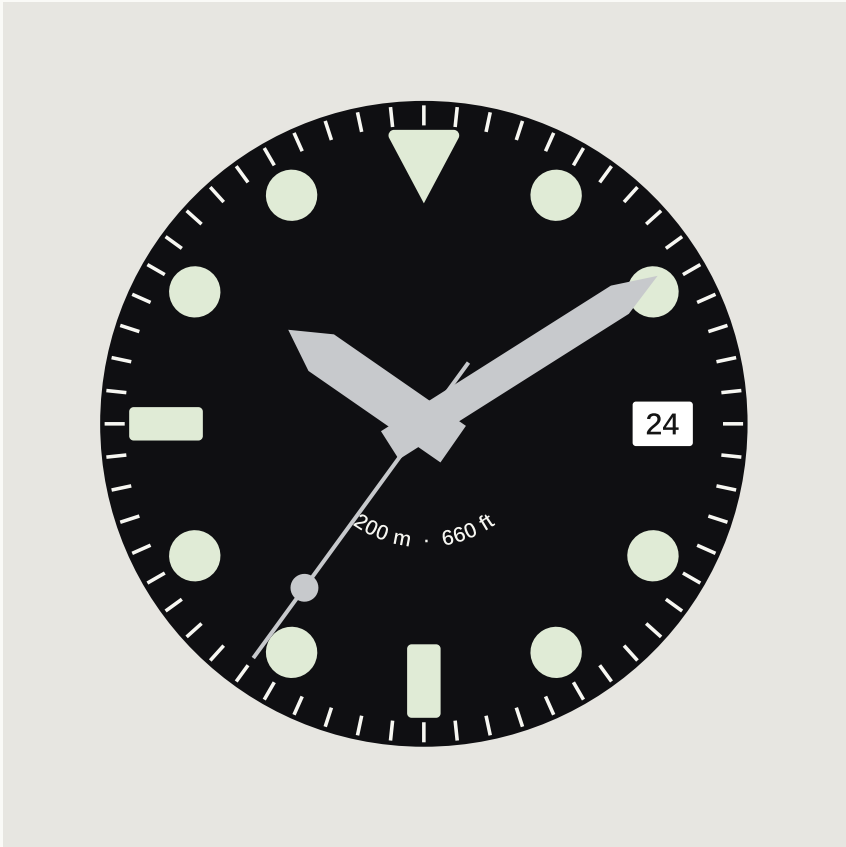
Diver

After the 1950s dive watch: fat dots, bars at 6 and 9, a triangle at 12, the date at 3.

LOOK FOR

Three markers rings share the hours: dots on **Every 3**, **skip those instead**; bars on Count 4 with the first position set to none; one triangle. The date takes 3 by itself.

presets/diver.json
Dial 29 mm ●



RINGS, TOP FIRST

☐ Triangle	markers
☒ Bars 6 · 9	markers
☒ Dots	markers
☒ Depth	numerals
☒ Minutes	ticks

SETTINGS THAT DIFFER FROM A NEW RING

Triangle markers	Radius 13.2 · Count 1 · Shape wedge · Length 3.3 · Width 3.5 · Inner width × 0 · Round outer 0.25 ●
Bars 6 · 9 markers	Radius 13.2 · Count 4 · Length 3.3 · Width 1.5 · Round outer 0.2 · Round inner 0.2 · 12 o'clock none ●
Dots markers	Radius 13 · Every 3 · Skip those instead · Shape dot · Ø 2.3 ●
Depth numerals	Radius 5.2 · Count 1 · Start 180° · Labels Custom · Custom 200 m · 660 ft · Placement on path · Size 0.95 · Font Helvetica Neue Medium ○
Minutes ticks	Radius 14.3 · Count 60 · Length 0.9 · Width 0.16 ○
Date window	Position 3 · Radius 10.7 · Width 2.7 · Height 2 · Corner 0.15 · Frame 0 · Print clearance 0.3 · Shows 24 · Text 68 %
Hour hand	Shape pencil · Length 7.4 · Width 2 · Tail 1.6 · Tip length 1.8 ●
Minute hand	Shape pencil · Length 12.4 · Width 1.5 · Tail 1.8 · Tip length 2 ●
Seconds hand	Shape lollipop · Length 13 · Width 0.18 · Tail 3.4 · Head · ring · disc 1.25 · at 70 % ●
Cap	Ø 1.6 ●

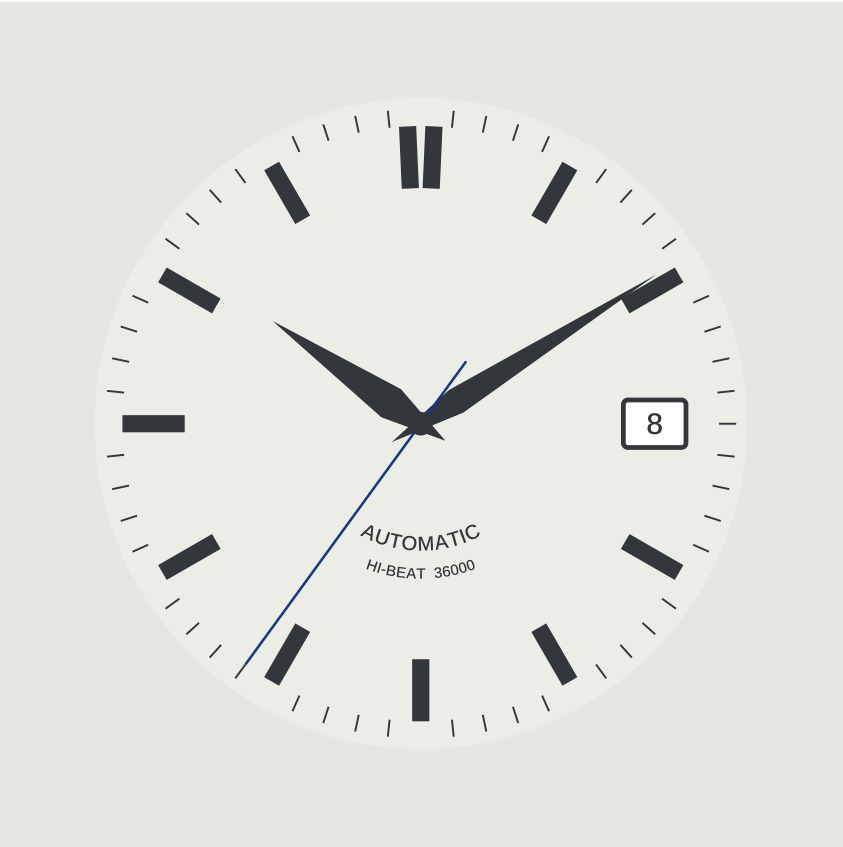
Dress watch

After the Japanese dress watches of the 1960s: plain bars, a double bar at 12, dauphine hands, a framed date.

LOOK FOR

The double bar is the markers ring's 12 o'clock style. Both lines of text are numerals rings with Count 1, set on path at 180°.

presets/dress.json
Dial 30.4 mm



RINGS, TOP FIRST

☐ Hours	markers
☒ Minutes	ticks
☒ Line 1	numerals
☒ Line 2	numerals

SETTINGS THAT DIFFER FROM A NEW RING

Hours markers	Radius 13.9 · Length 2.9 · Width 0.8 · 12 o'clock double · Gap 0.3 ●
Minutes ticks	Radius 14.7 · Count 60 · Length 0.8 · Width 0.09 · Skip where markers above ●
Line 1 numerals	Radius 5.6 · Count 1 · Start 180° · Labels Custom · Custom AUTOMATIC · Placement on path · Size 0.95 · Font Helvetica Neue Medium ●
Line 2 numerals	Radius 7 · Count 1 · Start 180° · Labels Custom · Custom HI-BEAT 36000 · Placement on path · Size 0.7 · Font Helvetica Neue Medium ●
Date window	Position 3 · Radius 10.9 · Width 2.7 · Height 2 · Corner 0.1 · Frame 0.22 · Print clearance 0.25 · Shows 8 · Text 70 %
Hour hand	Shape dauphine · Length 8.4 · Width 1.6 · Tail 1.4 · at 20 % ●
Minute hand	Shape dauphine · Length 13 · Width 1.25 · Tail 1.6 · at 15 % ●
Seconds hand	Shape baton · Length 13.9 · Width 0.12 · Tail 3.6 ●
Cap	Ø 1.1 ●

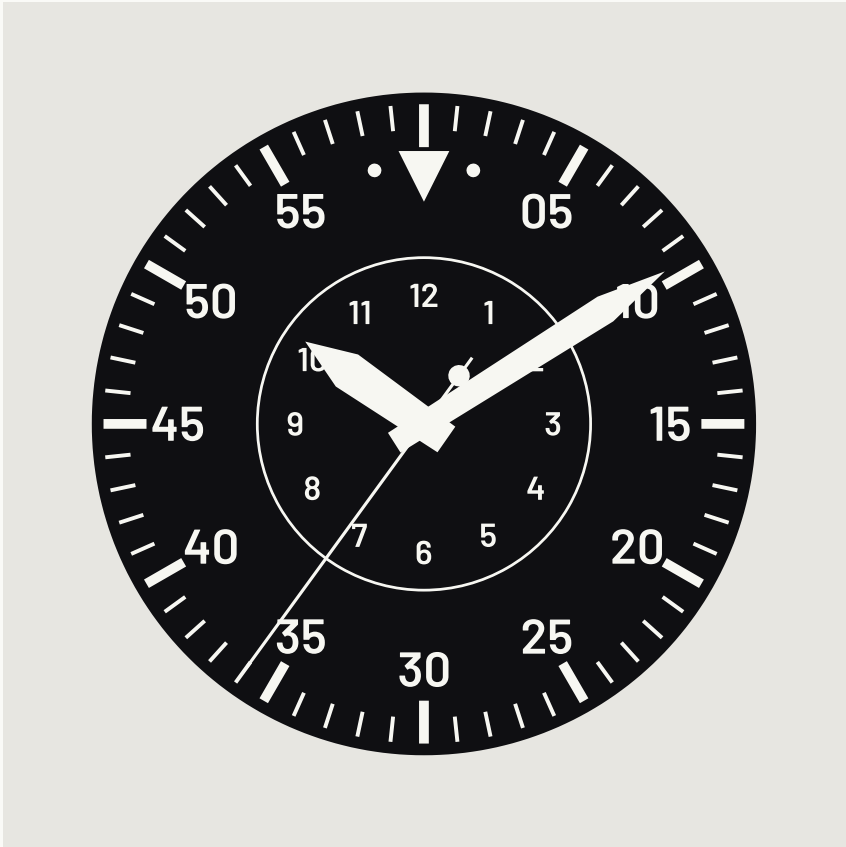
Pilot, type B

After the 1940s observer's watch: minutes outside, hours on an inner ring, a triangle and two dots at 12.

LOOK FOR

The minutes are **Numbers**: first 5, step 5, two digits, over a 300° span that leaves 12 free for the triangle. The two dots are a ticks ring with Count 2 on a 22° span.

presets/pilot.json
Dial 34 mm ●



RINGS, TOP FIRST

☐ Minutes	numerals
☒ Triangle	markers
☒ Two dots	ticks
☒ Five minutes	markers
☒ Minute track	ticks
☒ Hours	numerals
☒ Hour circle	band

SETTINGS THAT DIFFER FROM A NEW RING

Minutes numerals	Radius 12.6 · Count 11 · Start 30° · Span 300° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Size 2.5 · Font DIN Alternate Bold ○
Triangle markers	Radius 14 · Count 1 · Shape wedge · Length 2.6 · Width 2.6 · Inner width × 0 ○
Two dots ticks	Radius 13.6 · Count 2 · Start -11° · Span 22° · Shape dot · Ø 0.7 ○
Five minutes markers	Radius 16.4 · Length 2.2 · Width 0.5 ○
Minute track ticks	Radius 16.4 · Count 60 · Length 1.3 · Width 0.2 · Skip where markers above ○
Hours numerals	Radius 6.6 · Size 1.7 · Font DIN Alternate Bold ○
Hour circle band	Radius 8.6 · Band width 0.14 ○
Hour hand	Shape sword · Length 7.4 · Width 2 · Tail 1.4 · Tip length 2.6 ○
Minute hand	Shape sword · Length 14.6 · Width 1.5 · Tail 1.8 · Tip length 3.6 ○
Seconds hand	Shape baton · Length 15.4 · Width 0.16 · Tail 4.2 · Counterweight Ø 1.1 ○
Cap	Ø 1.7 ○

California

After the 1930s–40s dial:
Roman on top, Arabic below,
bars at 3 · 6 · 9, a triangle at
12, a railroad track.

LOOK FOR

Two numerals rings on one
radius, each with a **Custom**
list that leaves the other's
positions empty – so Roman
and Arabic can use different
fonts and placements.

presets/california.json
Dial 30 mm ●



RINGS, TOP FIRST

☐ Roman	numerals
☒ Arabic	numerals
☒ Bars + triangle	markers
☒ Minutes	ticks
☒ Outer rail	band
☒ Inner rail	band

SETTINGS THAT DIFFER FROM A NEW RING

Roman numerals	Radius 9.6 · Labels Custom · Custom , I, II, , , , , , , , X, XI · Placement radial · Size 3.3 · Font Bodoni 72 Book ●
Arabic numerals	Radius 9.6 · Labels Custom · Custom , , , , , 4, 5, , 7, 8, , , , · Placement upright · Size 3.2 · Font DIN Alternate Bold ●
Bars + triangle markers	Radius 11.4 · Count 4 · Length 3.4 · Width 1 · 12 o'clock triangle · Scale × 1.25 ●
Minutes ticks	Radius 14.2 · Count 60 · Length 1 · Width 0.14 ●
Outer rail band	Radius 14.2 · Band width 0.12 ●
Inner rail band	Radius 13.32 · Band width 0.12 ●
Hour hand	Shape pencil · Length 7.2 · Width 1.5 · Tail 1.4 · Tip length 1.6 ●
Minute hand	Shape pencil · Length 11.8 · Width 1.2 · Tail 1.6 · Tip length 1.8 ●
Seconds hand	Shape baton · Length 12.6 · Width 0.13 · Tail 3.2 · Counterweight Ø 0.9 ●
Cap	Ø 1.4 ●

Station clock

After the Swiss railway clock of 1944: heavy bars, and a red seconds hand like a dispatcher's baton.

LOOK FOR

Two rings. Everything is in the proportions: bars 3.7×1.3 , ticks 1.3×0.45 , and a lollipop with its disc at 100 % of the length.

presets/railway.json
Dial 30 mm 



RINGS, TOP FIRST

☐ Hours	markers
☒ Minutes	ticks

SETTINGS THAT DIFFER FROM A NEW RING

Hours markers	Radius 14.3 · Length 3.7 · Width 1.3 ●
Minutes ticks	Radius 14.3 · Count 60 · Length 1.3 · Width 0.45 · Skip where markers above ●
Hour hand	Shape baton · Length 8.8 · Width 1.7 · Tail 2.4 ●
Minute hand	Shape baton · Length 13.6 · Width 1.3 · Tail 2.8 ●
Seconds hand	Shape lollipop · Length 9.8 · Width 0.34 · Tail 3.6 · Head · ring · disc 2.7 · at 100 % ●
Cap	Ø 1 ●

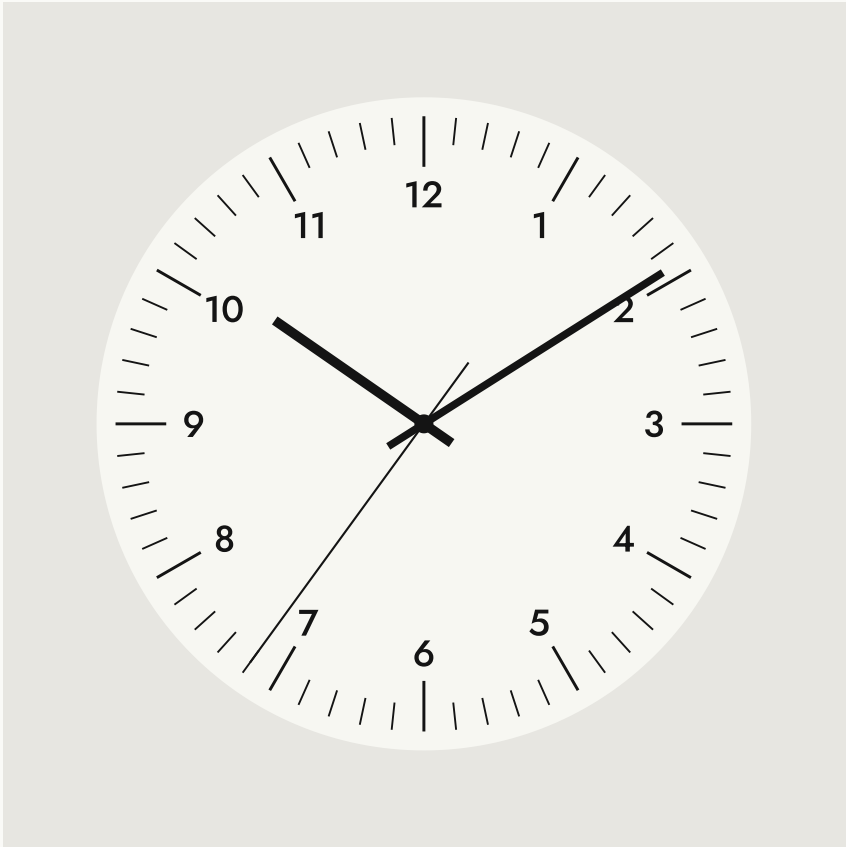
Bauhaus

After the German wall-clock school of the 1950s–60s: hairline marks, small numerals, thin hands.

LOOK FOR

Line weights carry it: 0.14 for the hours, 0.09 for the minutes, hands under half a millimetre.

presets/bauhaus.json
Dial 31 mm ○



RINGS, TOP FIRST

☐ Hours	numerals
☒ Hour lines	markers
☒ Minutes	ticks

SETTINGS THAT DIFFER FROM A NEW RING

Hours numerals	Radius 10.9 · Size 1.75 · Font Futura Medium ●
Hour lines markers	Radius 14.6 · Length 2.4 · Width 0.14 ●
Minutes ticks	Radius 14.6 · Count 60 · Length 1.3 · Width 0.09 · Skip where markers above ●
Hour hand	Shape baton · Length 8.6 · Width 0.46 · Tail 1.6 ●
Minute hand	Shape baton · Length 13.4 · Width 0.36 · Tail 2 ●
Seconds hand	Shape baton · Length 14 · Width 0.1 · Tail 3.6 ●
Cap	Ø 0.9 ●

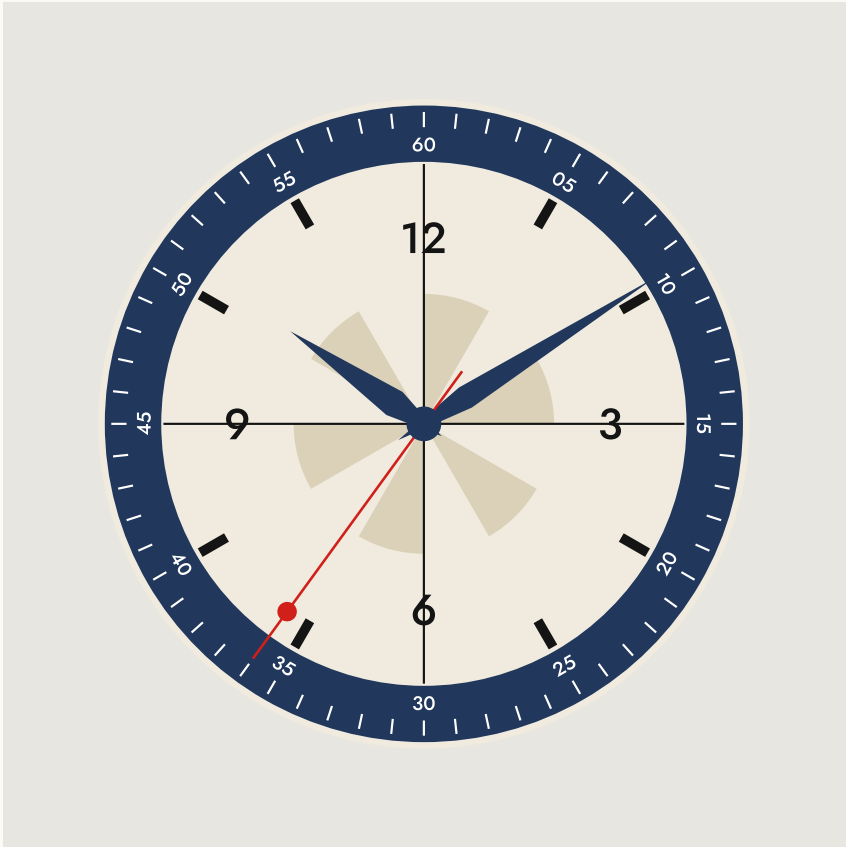
Sector dial

After the 1930s scientific dials: a chapter ring, a crosshair, zones of colour.

LOOK FOR

Bands do the colour: a chapter ring 2.6 wide, and pies made by Band width = Radius with Every 2. The crosshair is a ticks ring, Count 4, Length 12.

presets/sector.json
Dial 30 mm



RINGS, TOP FIRST

☐	Hours	numerals
☒	Hour bars	markers
☒	Minutes	ticks
☒	Minute numerals	numerals
☒	Chapter	band
☒	Crosshair	ticks
☒	Sectors	band

SETTINGS THAT DIFFER FROM A NEW RING

Hours numerals	Radius 8.6 · Every 3 · Size 2 · Font Futura Medium ●
Hour bars markers	Radius 11.9 · Every 3 · Skip those instead · Length 1.4 · Width 0.45 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 0.7 · Width 0.1 ○
Minute numerals numerals	Radius 12.9 · Start 30° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Placement radial, auto-flip · Size 0.9 · Font Futura Medium ○
Chapter band	Radius 14.7 · Band width 2.6 ●
Crosshair ticks	Radius 12 · Count 4 · Length 12 · Width 0.1 ●
Sectors band	Radius 6 · Count 12 · Every 2 · Band width 6 ○
Hour hand	Shape dauphine · Length 7.5 · Width 1.3 · Tail 1 · at 22 % ●
Minute hand	Shape dauphine · Length 12.6 · Width 1.1 · Tail 1.4 · at 18 % ●
Seconds hand	Shape lollipop · Length 13.4 · Width 0.12 · Tail 3 · Head · ring · disc 0.9 · at 80 % ●
Cap	Ø 1.6 ●

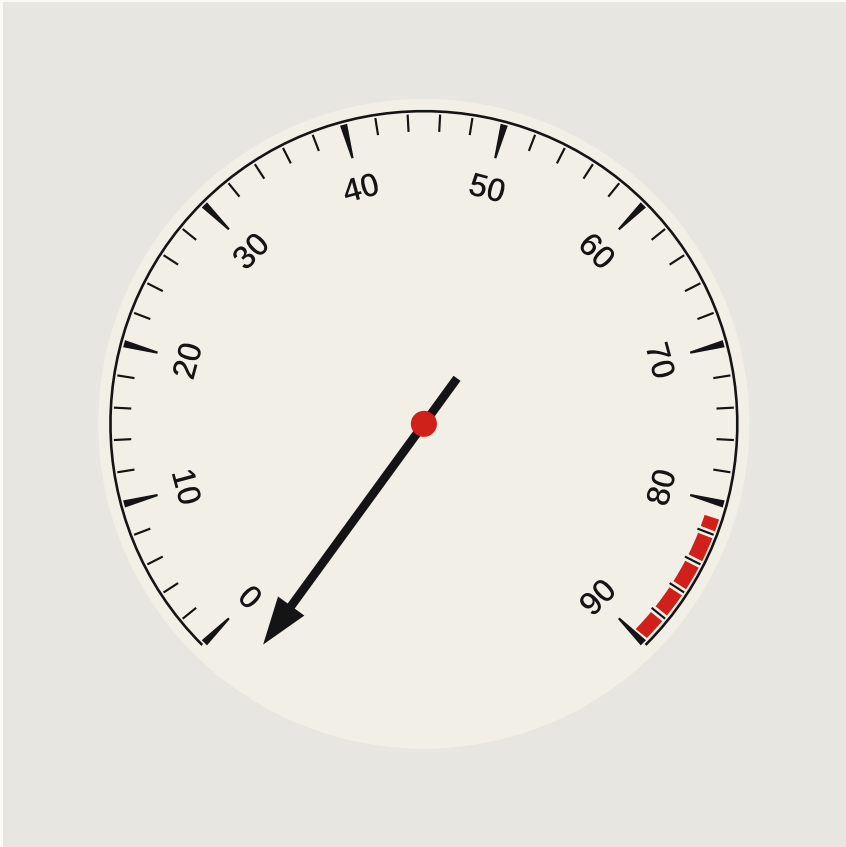
Gauge

A 270° scale with a red zone: positions over a partial span mark both ends.

LOOK FOR

Start -135°, Span 270°. A partial span marks both ends, so Count 10 gives 0 to 90. The red zone **knocks out** everything above it with a 0.12 clearance.

presets/gauge.json
Dial 30 mm 



RINGS, TOP FIRST

☐ Scale	numerals
☒ Scale marks	markers
☒ Scale ticks	ticks
☒ Rail	band
☒ Red zone	band

SETTINGS THAT DIFFER FROM A NEW RING

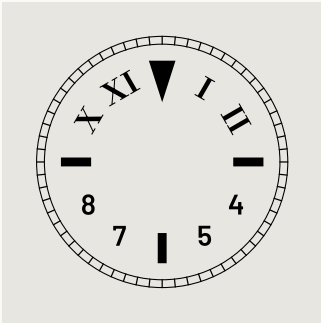
Scale numerals	Radius 11.3 · Count 10 · Start -135° · Span 270° · Labels Numbers · First number 0 · Step 10 · Placement radial, auto-flip · Size 1.5 · Font Helvetica Neue Medium ●
Scale marks markers	Radius 14.3 · Count 10 · Start -135° · Span 270° · Shape wedge · Length 1.6 · Width 0.34 · Inner width × 0.3 ●
Scale ticks ticks	Radius 14.3 · Count 46 · Start -135° · Span 270° · Length 0.8 · Width 0.1 · Skip where markers above ●
Rail band	Radius 14.5 · Count 1 · Start -135° · Span 270° · Band width 0.12 ●
Red zone band	Radius 14.5 · Count 1 · Start 108° · Span 27° · Band width 0.9 · Knock out everything above · Clearance 0.12 ●
Minute hand	Shape arrow · Length 12.6 · Width 0.4 · Tail 2.6 · Tip length 2.2 · Head · ring · disc 1.5 ●
Cap	Ø 1.2 ●

Files in and out

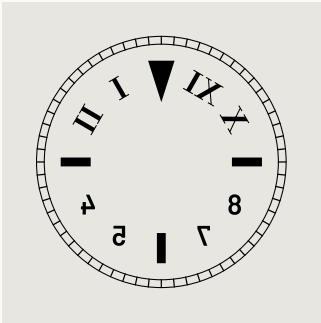
Vectors at true size, a one-colour mask for production, and your own shapes coming in.



design
PDF · SVG · PNG



production
one colour, no plate, no hands



production + mirror
for toner transfer

EXPORT

☒ Production: mono print mask

☒ Mirror: toner transfer

☐ Live text in PDF · SVG (fonts needed)

PNG dpi

Clip seconds

Frames per s

PDF · SVG

Vector, true size.

For Illustrator, print, the press, laser. A 30 mm dial opens 30 mm wide.

PNG

At your dpi.

Film for toner transfer. 30 mm at 600 dpi is 709 pixels; a 0.09 mm tick is two of them.

MP4 · GIF

Hands moving.

From the set time, for Clip seconds at Frames per s, in the chosen beat.

PRODUCTION

One colour.

No plate, no hands; every ring black; the date aperture a cut line. Mirror flips it for transfer.

LIVE TEXT

Off by default.

Keeps numerals editable in PDF and SVG. The fonts are then needed, and on-path text becomes one object per letter.

IN

SVG, PDF and AI, for markers and hands.

Draw pointing up, at 12 o'clock. The artboard's centre is the anchor: a hand's pivot, a marker's centre. Black is ink; white shapes cut holes; strokes become outlines.

BEFORE IMPORTING

Outline text, flatten effects.

Text, images and gradients are skipped, and the Export log says so. Save AI files with "Create PDF Compatible File". Only the first page counts.

SETTINGS

Save... Load...

A .json of everything. Defaults starts over.

Recipes Help

Short constructions to build on, with a dial that uses each. Below: what to check when something looks wrong.

TRACK Railroad. Ticks, Count 60. Two bands 0.12 wide: one at the ticks' Radius, one at Radius – tick Length + 0.12. Page 17.	TRACK Fifths of a second. Ticks Count 60 above ticks Count 300. The 300 Skip where: all marks above.	HOURS Cardinals only. Every 3. For the other eight: Every 3 and Skip those instead. Page 20.	HOURS 24-hour ring. Numerals, Labels Custom: 24, 13, 14 ... 23, at a smaller Radius and Size. Page 13.	HOURS Two fonts, one circle. Two numerals rings on one Radius; each Custom list leaves the other's positions empty. Page 17.								
MINUTES 05 to 60. Numerals, Labels Numbers, First number 5, Step 5, Digits 2, Start 30°. Page 20.	TEXT A wordmark. Numerals, Count 1, Start 0° or 180°, Labels Custom, Placement on path. Page 15.	COLOUR A gap round numerals. A band below the numerals; Knock out: markers & numerals above; Clearance 0.3.	COLOUR Sector dial. A band, Count 12, Every 2, Band width equal to Radius. Page 20.	LUME An opening. Hollow wall 0.15 to 0.25 on a marker or a hand leaves the outline as a frame.								
A RING IS GONE Three places to look. Its checkbox in the list; its Every and Offset; and whether it skips positions that a ring above takes.	NO NUMERAL AT 3 The date took it. Markers and numerals leave the date's position empty.	DIAMETER Nothing else moves. Radii and hand lengths are millimetres from the centre, as on a real movement. Change them ring by ring.	SLOW Check the log. It should say shape combining: skia-pathops. If not, install it: page 2.	START OVER Defaults. Export → Settings → Defaults. The last session is in ~/Library/Application Support/DialTool/last.json.								
WHEN SOMETHING IS WRONG Read the log, compare the previews, send both. 1 Open Export and read the log. At launch it says preview: canvas and all controls connected. 2 If the preview looks wrong, switch Canvas to PDF and compare. 3 Send a screenshot and the log. If the window doesn't build at all, run tools/dial_probe.py in DrawBot and send its report.												
WHERE THINGS LIVE One repository. <table><tr><td>Code • docs</td><td>github.com/duztman/dial-tool</td></tr><tr><td>Words</td><td>docs/glossary.md</td></tr><tr><td>Presets</td><td>docs/guide/presets/*.json</td></tr><tr><td>This guide</td><td>docs/guide/guide.html, built by tools/make_guide.py</td></tr></table>					Code • docs	github.com/duztman/dial-tool	Words	docs/glossary.md	Presets	docs/guide/presets/*.json	This guide	docs/guide/guide.html, built by tools/make_guide.py
Code • docs	github.com/duztman/dial-tool											
Words	docs/glossary.md											
Presets	docs/guide/presets/*.json											
This guide	docs/guide/guide.html, built by tools/make_guide.py											